

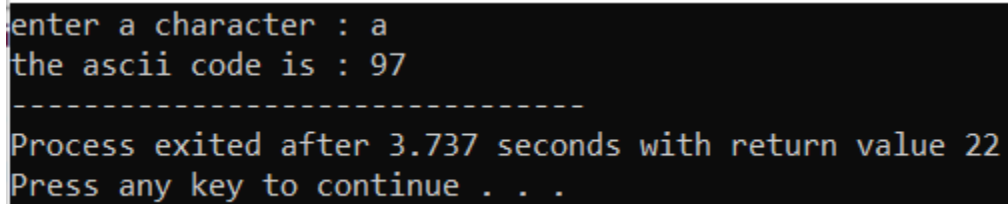
LAB 2:

Q-12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

Code:-

```
#include<stdio.h>
main()
{
    char ch;
    printf("enter a character : ");
    scanf("%c" , &ch);
    printf("the ascii code is : %d" , ch);
    return 0;
}
```

Output:-



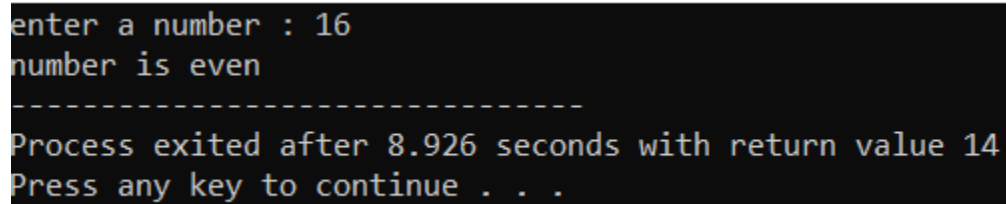
```
enter a character : a
the ascii code is : 97
-----
Process exited after 3.737 seconds with return value 22
Press any key to continue . . .
```

Q-13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

Code:-

```
#include<stdio.h>
main()
{
    int num;
    printf("enter a number : ");
    scanf("%d" , &num);
    if(num%2 == 0 && num!=0)
        printf("number is even");
    else if (num%2 != 0)
        printf("number is odd");
    else
        printf("entered number is zero");
    return 0;
}
```

Output:-

A screenshot of a terminal window showing the output of the program. The text is as follows:

```
enter a number : 16
number is even
-----
Process exited after 8.926 seconds with return value 14
Press any key to continue . . .
```

Q-14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

Code:-

```
#include<stdio.h>
main()
{
    char ch;
    printf("enter something");
    scanf("%c" , &ch );
    if(ch>='A' && ch<='Z')
        printf("you entered a capital letter");
    else if(ch>='a' && ch<='z')
        printf("you entered a small letter");
    else if(ch>='1' && ch<='10')
        printf("you entered a digit");
    else
        printf("you entered a special symbol");
    return 0;
}
```

Output:-

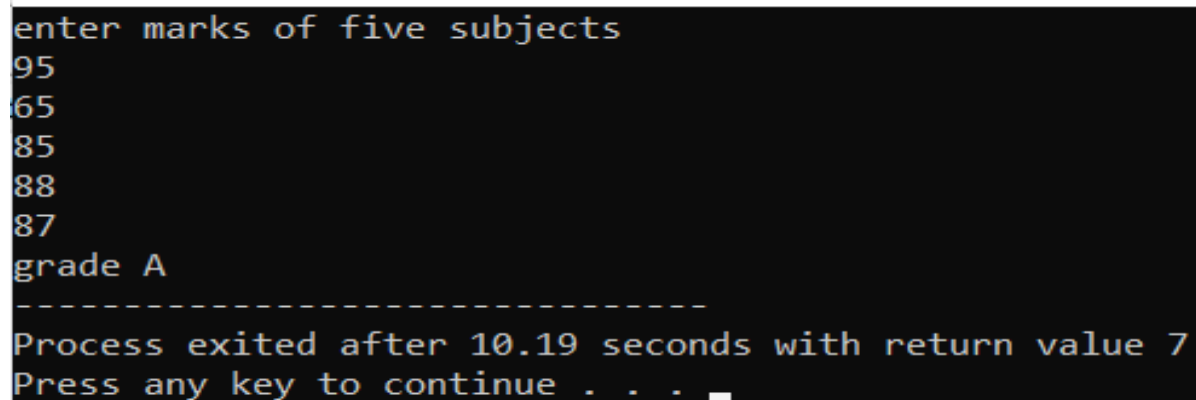
```
enter something*
you entered a special symbol
-----
Process exited after 6.556 seconds with return value 28
Press any key to continue . . .
```

Q-15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.

Code:-

```
#include<stdio.h>
main()
{
    int s1,s2,s3,s4,s5,total;
    printf("enter marks of five subjects \n");
    scanf("%d%d%d%d%d", &s1, &s2, &s3, &s4, &s5);
    total = (s1+s2+s3+s4+s5)/5;
    if(total>=90 && total<=100)
        printf("grade A+");
    else if(total>=70 && total<=89)
        printf("grade A");
    else if(total>=50 && total<=69)
        printf("grade B");
    else if(total>=30 && total<=49)
        printf("grade C");
    else
        printf("FAILED");
    return 0;
}
```

Output:-



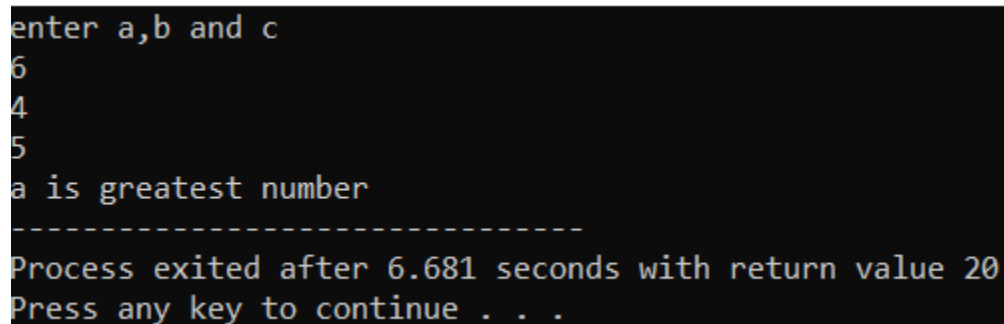
```
enter marks of five subjects
95
65
85
88
87
grade A
-----
Process exited after 10.19 seconds with return value 7
Press any key to continue . . . _
```

Q-16. PROGRAM TO READ THREE NOS. AND PRINT MAX.

Code:-

```
#include<stdio.h>
main()
{
    int a,b,c;
    printf("enter a,b and c\n");
    scanf("%d%d%d" , &a,&b,&c);
    if (a>b)
    { if(a>c)
      printf("a is greatest number");
    }
    else if (b>c)
    { if(b>a)
      printf("b is greatest number");
    }
    else
    printf("c is greatest");
    return 0;
}
```

Output:-



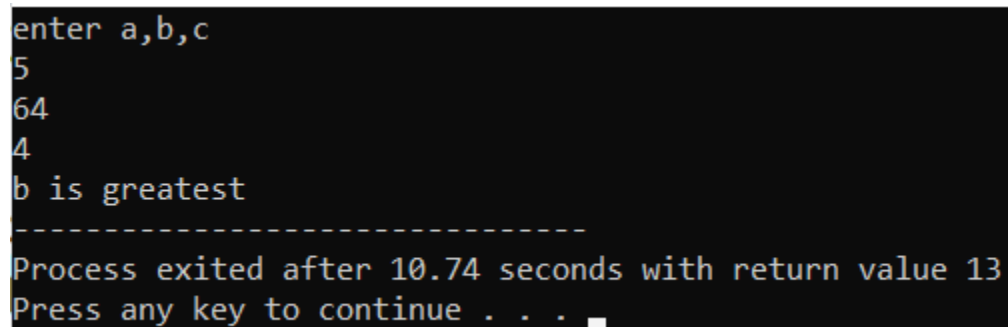
```
enter a,b and c
6
4
5
a is greatest number
-----
Process exited after 6.681 seconds with return value 20
Press any key to continue . . .
```

Q-17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

Code:-

```
#include<stdio.h>
main()
{
    int a,b,c;
    printf("enter a,b,c\n");
    scanf("%d%d%d" ,&a , &b,&c);
    if(a>b && a>c)
        printf("a is greatest");
    else if(b>c && b>a)
        printf("b is greatest");
    else
        printf("c is greatest");
    return 0;
}
```

Output:-



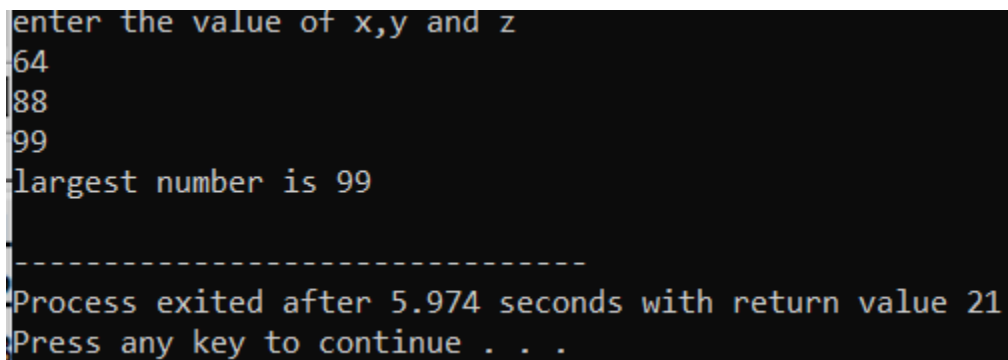
```
enter a,b,c
5
64
4
b is greatest
-----
Process exited after 10.74 seconds with return value 13
Press any key to continue . . .
```

Q-18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR.

Code:-

```
#include<stdio.h>
main()
{
    int x,y,z,max;
    printf("enter the value of x,y and z\n");
    scanf("%d%d%d", &x, &y , &z);
    max = ( x>y && x>z ? x : (y>z ? y : z));
    printf("largest number is %d\n" , max);
    return 0;
}
```

Output:-



```
enter the value of x,y and z
64
88
99
largest number is 99

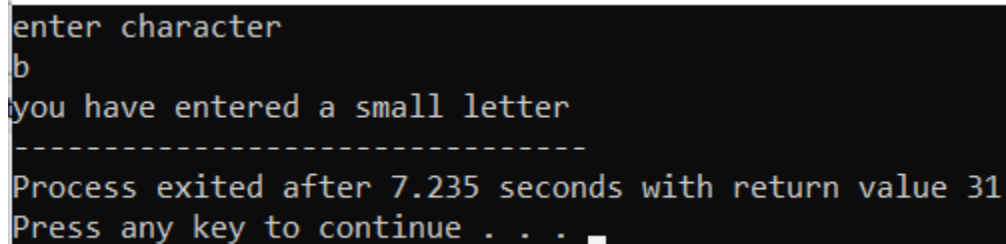
-----
Process exited after 5.974 seconds with return value 21
Press any key to continue . . .
```

Q-19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

Code:-

```
#include<stdio.h>
main()
{
    char ch;
    printf("enter character\n");
    scanf("%c" , &ch);
    if(ch>='a' && ch<='z')
        printf("you have entered a small letter");
    else
        printf("it is not a small letter");
}
```

Output:-



```
enter character
b
you have entered a small letter
-----
Process exited after 7.235 seconds with return value 31
Press any key to continue . . .
```

Q-20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

Code:-

```
#include<stdio.h>
main()
{
    float x, y;
    int command;
    printf("enter value of x and y\n");
    scanf("%f%f", &x, &y);
    printf("give command for addition = 1 , subtraction = 2 , multiplication = 3 and division = 4\n");
    scanf("%d", &command);
    switch(command){
        case 1 : printf("%f", x+y);
        break;
        case 2 : printf("%f", x-y);
        break;
        case 3 : printf("%f", x*y);
        break;
        case 4 : if(y==0)
        {
            printf("error");
        }
        else{
            printf("%f", x/y);
        } break;
        default: printf("invalid input");
    }
}
```

Output:-

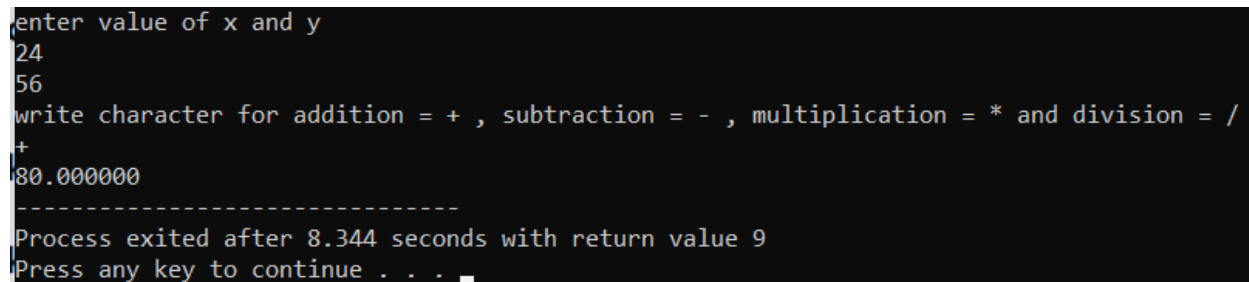
```
enter value of x and y
20
56
give command for addition = 1 , subtraction = 2 , multiplication = 3 and division = 4
3
1120.000000
-----
Process exited after 13.94 seconds with return value 11
Press any key to continue . . .
```

Q-21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR)AND THIS PERFORM USING SWITCH-CASE.(CASE IN CHAR FORM).

Code:-

```
#include<stdio.h>
main()
{
    float x, y;
    char command;
    printf("enter value of x and y\n");
    scanf("%f%f",&x,&y);
    printf("write character for addition = + , subtraction = - , multiplication = * and division = /\n");
    scanf(" %c",&command);
    switch(command)
    {
        case '+': printf("%f" , x+y);
        break;
        case '-': printf("%f" , x-y);
        break;
        case '*': printf("%f" , x*y);
        break;
        case '/': if(y==0)
        {
            printf("error");
        }
        else{
            printf("%f" , x/y);
        } break;
        default: printf("invalid input");
    }
}
```

Output:-



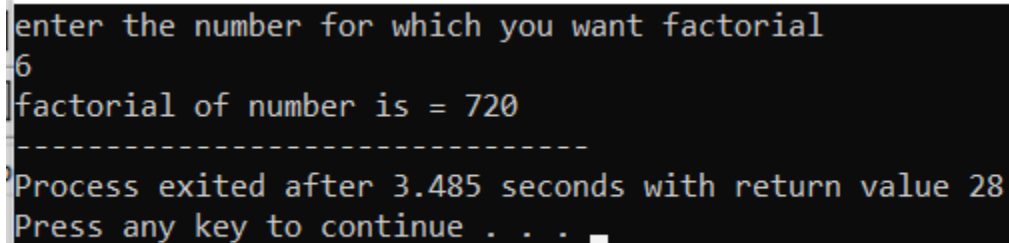
```
enter value of x and y
24
56
write character for addition = + , subtraction = - , multiplication = * and division = /
+
80.000000
-----
Process exited after 8.344 seconds with return value 9
Press any key to continue . . .
```

Q-22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

Code:-

```
#include<stdio.h>
main()
{
    int i,x=1,fac;
    printf("enter the number for which you want factorial\n");
    scanf("%d",&fac);
    for(i=1;i<=fac;i++)
    {
        x=x*i;
    }
    printf("factorial of number is = %d",x);
}
```

Output:-



```
enter the number for which you want factorial
6
factorial of number is = 720
-----
Process exited after 3.485 seconds with return value 28
Press any key to continue . . .
```

Q-23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

Code:-

```
#include<stdio.h>
main()
{
    float base,power,i,multiplier;
    printf("enter the value of base and the power\n");
    scanf("%f%f",&base,&power);
    multiplier=1;
    if(base!=0 && power>0)
    {
        for(i=1;i<=power;i++)
        {
            multiplier=multiplier*base;
        }
        printf("%f",multiplier);
    }
    else if(base!=0 && power<0){
    {
        for(i=1;i<=(-power);i++)
        multiplier=multiplier/base;
        printf("%f",multiplier);
    }
    }
    else
        printf("%f",multiplier);
}
```

Output:-

```

enter the value of base and the power
5
3
125.000000
-----
Process exited after 2.462 seconds with return value 10
Press any key to continue . . .

```

Q-24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

$$Y(x,n) = \begin{array}{ll} 1+x & \text{where } n=1 \\ 1+x/n & \text{where } n=2 \\ 1 + xn & \text{when } n=3 \\ 1 + nx & \text{when } n>3 \text{ or } n<1 \end{array}$$

Code:-

```

#include<stdio.h>
main()
{
    float x,n,y;
    printf("enter the value of x and n\n");
    scanf("%f%f",&x,&n);
    if(n==1)
    {
        y=1+x;
        printf("value of y is: %f",y);
    }
    else if(n==2)
    {
        y=1+(x*0.5);
        printf("value of y is: %f",y);
    }
    else if(n==3)
    {
        y=1+(x*x*x);
        printf("value of y is: %f",y);
    }
}

```

```
    else
    {
        y=1+(n*x);
        printf("value of y is :%f",y);
    }
}
```

Output:-

```
enter the value of x and n
2
2
value of y is: 2.000000
-----
Process exited after 11.76 seconds with return value 23
Press any key to continue . . .
```

Q-25. PROGRAM TO PRINT TABLE OF ANY NO.

Code:-

```
#include<stdio.h>
int main()
{
    int number,i,x;
    printf("enter a number for which you want to print table\n");
    scanf("%d",&number);
    for(i=1;i<=10;i++)
    {
        x=number*i;
        printf("%d*%d=%d\n",number,i,x);
    }
}
```

Output:-

```
enter a number for which you want to print table
8
8*1=8
8*2=16
8*3=24
8*4=32
8*5=40
8*6=48
8*7=56
8*8=64
8*9=72
8*10=80

-----
Process exited after 2.449 seconds with return value 8
Press any key to continue . . .
```

Q-26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

Code:-

```
#include<stdio.h>
main()
{
    int i,N,sum_odd=0,sum_even=0,sum;
    printf("enter the value N\n");
    scanf("%d",&N);
    for(i=1;i<=N;i++)
    {
        if(i%2!=0)
            sum_odd=sum_odd+i;
    }
    printf("sum off all odd integer between 1 to N is : %d\n",sum_odd);
    for(i=0;i<=N;i++)
    {
```

```
        if(i%2==0)
            sum_even=sum_even+i;
    }
    printf("sum off all even integer between 1 to N is : %d",sum_even);
}
```

Output:-

```
enter the value N
16
sum off all odd integer between 1 to N is : 64
sum off all even integer between 1 to N is : 72
-----
Process exited after 9.579 seconds with return value 47
Press any key to continue . . .
```